

Quantitative Risk Management Report

Detailed interpretation of the models, tables, and figures in
`Quant_Risk_Mgmt_project.ipynb`

Report objective. This report turns the saved notebook output into a submission-ready narrative. Section 1 explains every major risk model and equation used in the notebook, including how to interpret the quantities in practice. Section 2 walks through every tabular result and graph in a way that is meant to be readable even for someone who has just joined a quant risk team. Section 3 gives the overall conclusions, and Section 4 collects the derivations.

Portfolio value	USD 10,000,000
Assets	AAPL, MSFT, JPM, XOM, SPY, TLT, GLD, QQQ, IWM, LQD
Sample start	2007-01-01
Last visible date in saved run	2026-03-06
Estimated EWMA lambda	0.941929
EWMA half-life	11.59 trading days
Approximate effective memory	17.22 observations

Prepared from the stored notebook outputs

March 9, 2026

Contents

1	Introduction and sign convention	3
2	Models, equations, properties, and interpretation	3
2.1	Data construction and portfolio P&L	3
2.2	Value-at-Risk and Expected Shortfall	4
2.3	Historical simulation	4
2.4	Bootstrap historical simulation	5
2.5	Exponentially weighted historical simulation	5
2.6	EWMA volatility model and volatility-weighted historical simulation	6
2.7	Correlation-weighted historical simulation	6
2.8	Delta-normal VaR	7
2.9	Monte Carlo simulation	7
2.10	Risk contributions	8
2.11	Backtesting: Kupiec and Christoffersen tests	8
2.12	Scenario analysis and historical stress testing	8
3	Results, tables, graphs, and practical interpretation	9
3.1	Portfolio setup and data pipeline outputs	9
3.2	Baseline historical simulation	14
3.3	Bootstrap historical simulation	15
3.4	Exponentially weighted historical simulation	22
3.5	Volatility-weighted historical simulation	26
3.6	Correlation-weighted historical simulation	27
3.7	VaR backtesting	31
3.8	Scenario analysis	34
3.9	Historical stress testing	35
4	Summary	36
5	Appendix: derivations	37
5.1	From price to P&L	37
5.2	Historical-simulation VaR and ES	37
5.3	Weighted quantile and weighted ES	38
5.4	EWMA likelihood and lambda estimation	38

5.5	Volatility scaling derivation	38
5.6	Correlation adjustment derivation	38
5.7	Delta-normal VaR and ES derivation	39
5.8	Kupiec test derivation	39
5.9	Christoffersen independence and conditional coverage	39
5.10	Why stress numbers exceed ordinary daily VaR	40

1 Introduction and sign convention

The notebook studies daily market risk for a static ten-asset portfolio using a family of Value-at-Risk (VaR) and Expected Shortfall (ES) methods. The workflow is straightforward: adjusted close prices are downloaded, converted into daily log returns and simple returns, combined into a portfolio return series, and then transformed into daily profit-and-loss (P&L). All subsequent risk numbers are different ways of summarizing the left tail of that P&L series.

Important sign convention note. The saved notebook mixes two conventions: (i) for most VaR results, the number is reported as a *positive loss amount*, such as “VaR 99 = 279,855.40”; (ii) for most ES results, the number is reported as the *signed average tail P&L*, which is negative in loss states; (iii) in the exponentially weighted historical simulation (EWHs) cell, the raw VaR itself is printed as a negative tail P&L quantile. To keep the report faithful to the notebook while still being easy to interpret, the original notebook values are shown in the tables, and the discussion converts them into positive loss magnitudes whenever needed.

A beginner should keep four questions in mind while reading any VaR report:

1. What is the horizon and confidence level?
2. Is the number a *quantile* (VaR) or an *average of worse outcomes* (ES)?
3. Is the result expressed as a positive loss amount, or as a signed P&L?
4. Has the model been backtested and stress-tested, or is it only an in-sample description?

2 Models, equations, properties, and interpretation

2.1 Data construction and portfolio P&L

Let $P_{i,t}$ denote the adjusted close price of asset i at date t . The notebook computes daily log returns as

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \quad (1)$$

and simple returns as

$$R_{i,t} = e^{r_{i,t}} - 1. \quad (2)$$

With portfolio weights $\mathbf{w} = (w_1, \dots, w_n)^\top$ that sum to one, the daily portfolio return is

$$R_{p,t} = \mathbf{w}^\top \mathbf{R}_t, \quad (3)$$

and for portfolio value V_0 the daily portfolio P&L is

$$\Pi_t = V_0 R_{p,t}. \quad (4)$$

How to interpret this. Every risk measure in the notebook is ultimately a statement about the empirical distribution of Π_t . If the left tail of Π_t is fat, skewed, or clustered in crisis periods, those features will show up in the risk metrics.

Key properties.

- Log returns are additive through time, which is convenient for filtering and covariance modeling.
- Simple returns are more intuitive for P&L because multiplying by V_0 directly produces dollar gains and losses.
- The portfolio is static in the notebook, so there is no rebalancing or transaction-cost layer.

2.2 Value-at-Risk and Expected Shortfall

For confidence level c and tail probability $\alpha = 1 - c$, the notebook's positive-loss VaR convention can be written as

$$\text{VaR}_c = -Q_\alpha(\Pi), \quad (5)$$

where $Q_\alpha(\Pi)$ is the α -quantile of the P&L distribution. At $c = 99\%$, VaR_{99} is the loss threshold that should only be exceeded on roughly 1 out of 100 trading days under the model.

Expected Shortfall goes one step further and averages what happens *beyond* the VaR cutoff:

$$\text{ES}_c^{(\text{signed P\&L})} = \mathbb{E}[\Pi \mid \Pi \leq Q_\alpha(\Pi)]. \quad (6)$$

If one wants ES in positive loss language, the natural conversion is

$$\text{ES}_c^{(\text{loss})} = -\text{ES}_c^{(\text{signed P\&L})}. \quad (7)$$

How to interpret this. VaR answers “Where does the bad tail begin?” ES answers “Once I am already in the bad tail, how bad is the average outcome?” For a risk manager, VaR is a threshold metric; ES is a severity metric.

Key properties.

- VaR is easy to communicate, but it says nothing about the size of losses beyond the cutoff.
- ES is tail-sensitive and is a coherent risk measure, while VaR is not coherent in full generality because subadditivity can fail.
- For a properly specified loss distribution, ES in loss magnitude is typically at least as large as VaR at the same confidence level.

2.3 Historical simulation

Historical simulation (HS) is the simplest non-parametric method in the notebook. It uses the empirical distribution of past portfolio P&L directly:

$$\widehat{\text{VaR}}_c^{HS} = -Q_\alpha(\Pi_1, \dots, \Pi_T), \quad (8)$$

$$\widehat{\text{ES}}_c^{HS} = \frac{1}{N_\alpha} \sum_{t: \Pi_t \leq Q_\alpha} \Pi_t, \quad (9)$$

where N_α is the number of tail observations.

How to interpret this. HS says: “If the future looks like the past sample, what loss threshold do the worst α fraction of observations imply?” It preserves skewness, fat tails, and nonlinear cross-asset behavior because it never forces the data into a parametric distribution.

Key properties.

- No distributional assumption.
- Sensitive to sample choice and window length.
- Can be slow to react to regime shifts because all observations receive equal weight.
- Extreme risk depends heavily on whether the sample contains crisis episodes.

2.4 Bootstrap historical simulation

Bootstrap historical simulation does not create a new risk model for the market. Instead, it resamples the observed P&L series with replacement to measure the estimation uncertainty of the HS VaR and ES:

$$\Pi_1^{*(b)}, \dots, \Pi_m^{*(b)} \sim \text{resample with replacement from } \{\Pi_t\}_{t=1}^T, \quad (10)$$

and for each bootstrap replication b , VaR and ES are recomputed. The resulting bootstrap distribution of $\widehat{\text{VaR}}$ or $\widehat{\text{ES}}$ can then be summarized by a mean, standard deviation, and confidence interval.

How to interpret this. Bootstrap answers a different question from HS. It does *not* ask “What is market risk?” It asks “How much sampling uncertainty is there in my estimate of market risk?”

Key properties.

- Useful for confidence intervals and estimator stability.
- Works best when many bootstrap replications are used.
- Tail metrics at 99% are naturally less stable than at 95% because fewer observations live in the extreme tail.
- If the original historical sample is unrepresentative, bootstrap will inherit that weakness.

2.5 Exponentially weighted historical simulation

EWHS keeps the historical-simulation idea but assigns larger probability to recent observations. For a sample of length T , the notebook defines weights

$$w_t = \frac{(1 - \lambda)\lambda^{T-t}}{\sum_{s=1}^T (1 - \lambda)\lambda^{T-s}}, \quad (11)$$

so recent observations receive much larger weight than distant observations. The weighted VaR is then a weighted quantile of the historical P&L sample, and the weighted ES is a weighted average over the tail.

How to interpret this. EWHS says: “The past still matters, but last month matters more than last decade.” That makes it more responsive to current conditions, but it can also make old crises nearly irrelevant.

Key properties.

- Recency-sensitive and fast-reacting.

- The memory parameter λ controls how quickly history is forgotten.
- With $\hat{\lambda} \approx 0.9419$, the half-life is only about 11.6 trading days, so weights decay very quickly.
- If the implementation uses inconsistent sign conventions, backtesting can be badly distorted; this matters later in Section 2.

2.6 EWMA volatility model and volatility-weighted historical simulation

The notebook estimates an EWMA volatility model:

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda)(r_{t-1} - \mu)^2. \quad (12)$$

It then estimates λ by maximum likelihood. Once a volatility path is available, the notebook rescales each historical return so that all observations are expressed in today's volatility regime:

$$\tilde{r}_{i,t} = \mu_i + (r_{i,t} - \mu_i) \frac{\sigma_{i,T}}{\sigma_{i,t}}. \quad (13)$$

The rescaled returns are then transformed back into simple returns and portfolio P&L, and HS is run on that filtered series.

How to interpret this. Volatility-weighted HS tries to answer: “What would historical shocks look like if they happened under today's volatility level?” A large old shock that occurred during an already volatile regime may be toned down; a moderate old shock from a quiet regime may be scaled up.

Key properties.

- Still data-driven like HS, but more adaptive to current volatility.
- Often called filtered historical simulation.
- Usually more responsive than plain HS during changing volatility regimes.
- Remains sensitive to the correctness of the volatility filter.

2.7 Correlation-weighted historical simulation

The notebook also adjusts dependence structure, not just individual volatilities. It computes an EWMA covariance matrix

$$\mathbf{S}_t = \lambda \mathbf{S}_{t-1} + (1 - \lambda) \mathbf{x}_t \mathbf{x}_t^\top, \quad (14)$$

where $\mathbf{x}_t$ is the mean-adjusted return vector. Let $\mathbf{L}_t$ be the Cholesky factor of $\mathbf{S}_t$, and $\mathbf{L}_T$ the Cholesky factor of the current covariance matrix. The notebook maps each historical shock into a current-correlation shock by

$$\mathbf{z}_t = \mathbf{L}_t^{-1}(\mathbf{r}_t - \boldsymbol{\mu}), \quad \tilde{\mathbf{r}}_t = \boldsymbol{\mu} + \mathbf{L}_T \mathbf{z}_t. \quad (15)$$

How to interpret this. This method asks: “What would the old shock have looked like if it had occurred with today's covariance and correlation structure?” It is especially useful when the current co-movement of assets differs meaningfully from the unconditional historical average.

Key properties.

- Can materially change portfolio risk when current correlations tighten or loosen.
- Useful for multi-asset portfolios where diversification is correlation-dependent.
- Can become procyclical: if current correlations look benign, risk estimates can fall substantially.
- More model-dependent than plain HS because a covariance filter sits inside the procedure.

2.8 Delta-normal VaR

The notebook defines a parametric delta-normal VaR as if portfolio returns are approximately normal:

$$\mu_p = \mathbf{w}^\top \boldsymbol{\mu}, \quad \sigma_p = \sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}, \quad (16)$$

and for $z_c = \Phi^{-1}(c)$,

$$\text{VaR}_c^{DN} = V_0(z_c \sigma_p - \mu_p). \quad (17)$$

The associated ES formula under normality is

$$\text{ES}_c^{DN} = V_0 \left(\frac{\phi(z_c)}{1-c} \sigma_p - \mu_p \right). \quad (18)$$

How to interpret this. Delta-normal VaR replaces the empirical return distribution with a Gaussian approximation summarized by mean and covariance. It is fast, transparent, and analytically convenient, but it can underestimate risk when returns are skewed, heavy-tailed, or strongly regime-dependent.

Key properties.

- Very efficient computationally.
- Easy to decompose into marginal and component contributions.
- Best suited to near-linear portfolios and reasonably elliptical return distributions.
- Often challenged by empirical data during crisis periods.

2.9 Monte Carlo simulation

The notebook also defines a Monte Carlo VaR engine. It draws standard normal vectors $\mathbf{z}^{(s)}$, maps them through the Cholesky factor of the return covariance matrix, and creates simulated returns

$$\mathbf{R}^{(s)} = \boldsymbol{\mu} + \mathbf{L}\mathbf{z}^{(s)}, \quad (19)$$

then simulated portfolio returns and P&L. VaR and ES are finally estimated from the simulated P&L sample exactly as in HS.

How to interpret this. Monte Carlo is best thought of as a flexible scenario generator. In the notebook it uses Gaussian shocks, so it is still connected to the covariance structure of the data, but the framework can be extended to richer processes.

Key properties.

- Flexible and scalable.

- Separates modeling from numerical estimation.
- Sensitive to the choice of simulation model, not only to the number of simulations.
- In the saved notebook output, Monte Carlo is defined but not shown as a separate result table.

2.10 Risk contributions

The notebook includes delta-normal marginal and component VaR contributions. If portfolio volatility is σ_p , the marginal VaR contribution of asset i is

$$mVaR_i = V_0 \left(z_c \frac{(\Sigma \mathbf{w})_i}{\sigma_p} - \mu_i \right), \quad (20)$$

and the component contribution is

$$cVaR_i = w_i mVaR_i. \quad (21)$$

How to interpret this. These quantities tell the analyst which positions are responsible for the portfolio-level VaR, not merely which positions are volatile on a stand-alone basis. A low-volatility asset can still contribute strongly to portfolio risk if it loads heavily on common factors.

2.11 Backtesting: Kupiec and Christoffersen tests

Backtesting evaluates whether a VaR model behaves correctly out of sample. If I_t is the breach indicator, where $I_t = 1$ when realized P&L is worse than the VaR threshold and 0 otherwise, Kupiec's test checks whether the breach frequency matches the target tail probability α . Christoffersen's test adds the question of independence: are breaches scattered over time, or do they cluster?

How to interpret this. A model can pass the frequency test but still fail the independence test. That is exactly what happens in many practical portfolios: the model gets the average number of exceptions roughly right, but breaches arrive in bunches during crises, which means the model does not fully capture regime dependence.

Key properties.

- Kupiec tests calibration of counts.
- Christoffersen tests both counts and clustering.
- Low p-values indicate evidence against the model.
- Backtesting is indispensable because in-sample fit alone does not validate a risk model.

2.12 Scenario analysis and historical stress testing

VaR and ES are probabilistic summaries. Scenario analysis and stress testing answer different and complementary questions.

A scenario analysis imposes user-chosen shocks on asset classes and computes the resulting portfolio P&L:

$$\Pi^{scenario} = V_0 \mathbf{w}^\top \mathbf{s}, \quad (22)$$

where $\mathbf{s}$ is the vector of scenario returns.

Historical stress testing restricts attention to known crisis windows and recomputes tail metrics within those subperiods. This tells us how bad the portfolio would have looked in specific episodes such as the Global Financial Crisis or the COVID shock.

How to interpret this. Scenarios are “what-if” questions designed by the analyst. Historical stress tests are “what-happened” questions based on observed crisis windows. Together they prevent VaR from being mistaken for a full description of tail risk.

3 Results, tables, graphs, and practical interpretation

3.1 Portfolio setup and data pipeline outputs

Table 1: Portfolio composition used in the notebook.

Ticker	Asset class	Weight
AAPL	Equity	10.0%
MSFT	Equity	10.0%
JPM	Equity	10.0%
XOM	Equity	8.0%
SPY	Equity ETF	12.0%
TLT	Bond ETF	10.0%
GLD	Commodity ETF	8.0%
QQQ	Equity ETF	12.0%
IWM	Equity ETF	10.0%
LQD	Credit ETF	10.0%

The portfolio is heavily tilted toward equities and equity ETFs. Roughly 72% of the total weight is in equity-like exposures once individual stocks and equity ETFs are combined. That matters immediately for interpretation: equity selloffs should dominate the risk story, while TLT, LQD, and GLD are the main diversification sleeves.

Table 2: Last five adjusted-close observations shown in the notebook output.

Date	AAPL	MSFT	JPM	XOM	SPY	TLT	GLD	QQQ	IWM	LQD
2026-03-02	264.72	398.55	297.56	154.22	686.38	89.61	490.00	608.09	263.81	110.92
2026-03-03	263.75	403.93	300.26	151.83	680.33	89.43	468.14	601.58	259.24	110.87
2026-03-04	262.52	405.20	299.39	149.82	685.13	89.15	471.80	610.75	261.76	110.97
2026-03-05	260.29	410.68	293.55	150.76	681.31	88.79	466.13	608.91	256.76	110.53
2026-03-06	257.46	408.96	289.48	151.21	672.38	88.46	473.51	599.75	250.89	110.16

The last visible adjusted-close snapshot in the saved notebook run shows the portfolio data extending through 2026-03-06. The table is not a risk result by itself, but it confirms two practical points. First, the portfolio combines high-beta equity exposures with diversifiers. Second, the scale differences across names remind the analyst that risk comes from returns and covariance, not from comparing absolute price levels.

Table 3: Last five portfolio observations shown in the notebook output.

Date	portfolio_return	portfolio_pnl	portfolio_loss
2026-03-02	0.2509%	25,086.94	-25,086.94
2026-03-03	-0.7239%	-72,385.66	72,385.66
2026-03-04	0.2550%	25,496.94	-25,496.94
2026-03-05	-0.5648%	-56,482.39	56,482.39
2026-03-06	-0.7758%	-77,577.81	77,577.81

The last five portfolio observations show daily portfolio returns in the range of roughly -0.78% to $+0.26\%$. For a USD 10 million book, even seemingly small daily returns translate into five-figure and low six-figure P&L moves. This is a useful practical anchor: VaR numbers in the rest of the notebook are not abstract statistics; they sit on top of daily P&L moves of exactly this kind.

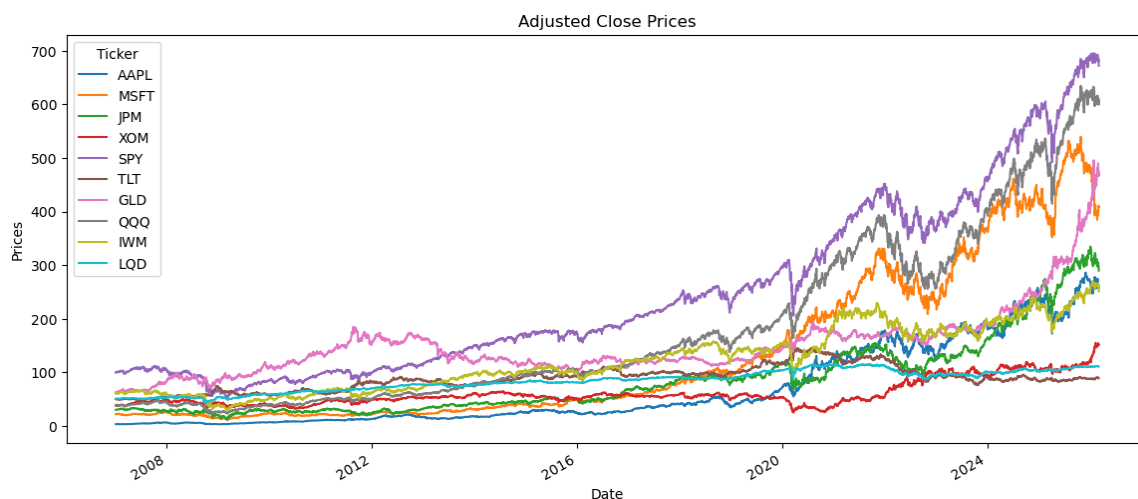


Figure 1: Adjusted close prices for all ten assets.

Interpretation. The price chart shows long-run growth in the equity and equity-ETF sleeve, with visibly different trend and volatility regimes across the sample. The bond and credit ETF sleeve is far more stable in price level, while gold follows its own path. For a new analyst, the key lesson is that a diversified portfolio does not mean every constituent behaves similarly; diversification comes from differences in return behavior, not from similar-looking price charts.

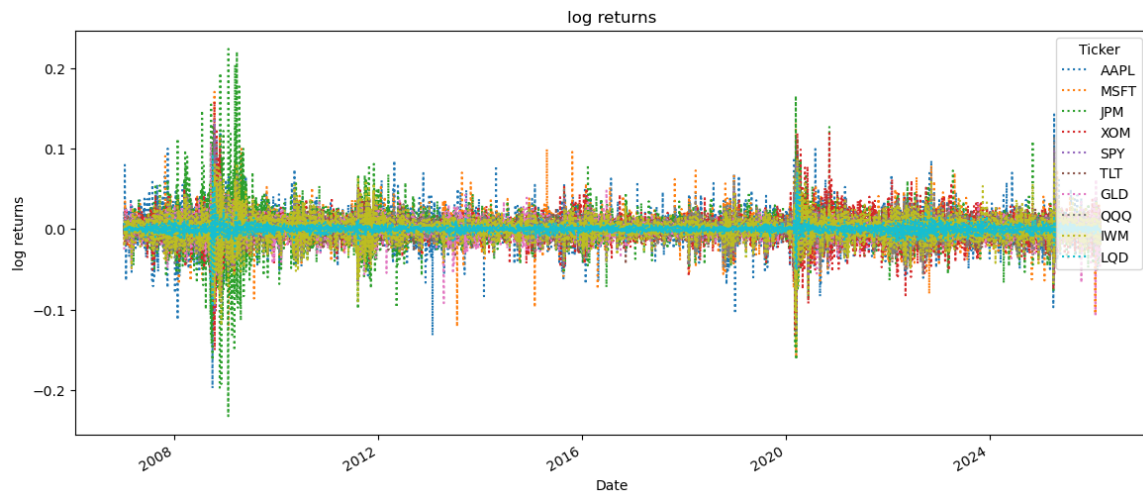


Figure 2: Daily log returns for all ten assets.

Interpretation. The log-return panel immediately reveals volatility clustering. The returns are mostly centered around zero, but the amplitude of the spikes changes dramatically over time. This is one of the clearest reasons plain normal models struggle in risk management: the variance is not constant. You can already see crisis periods where the distribution becomes much wider and more dangerous.

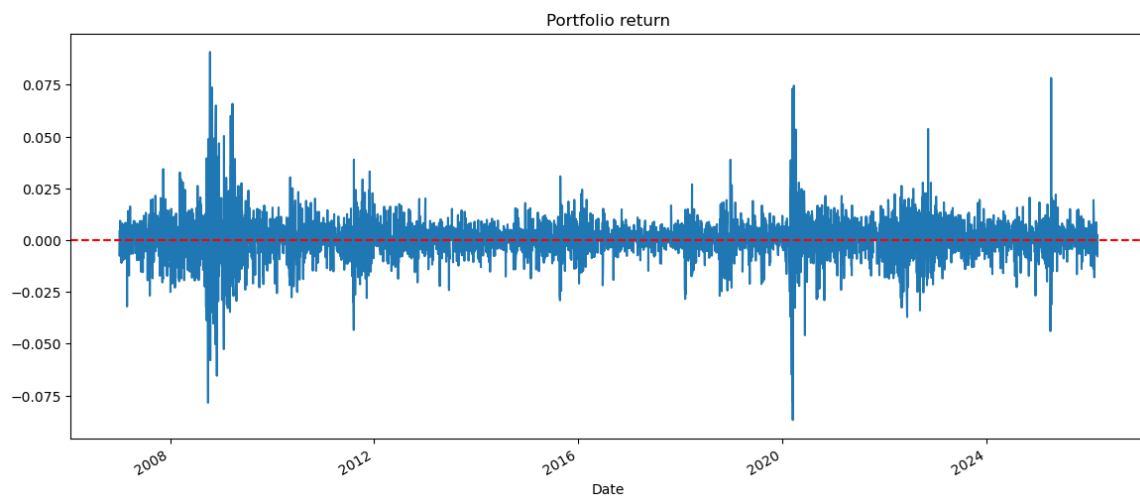


Figure 3: Daily portfolio return series with zero line.

Interpretation. The portfolio return series compresses all ten assets into one object. The red zero line helps distinguish normal oscillation from severe drawdown days. Large negative spikes are concentrated in a handful of stress windows instead of being spread evenly over time. That clustering is the reason backtesting later flags independence problems even when the average number of breaches looks acceptable.

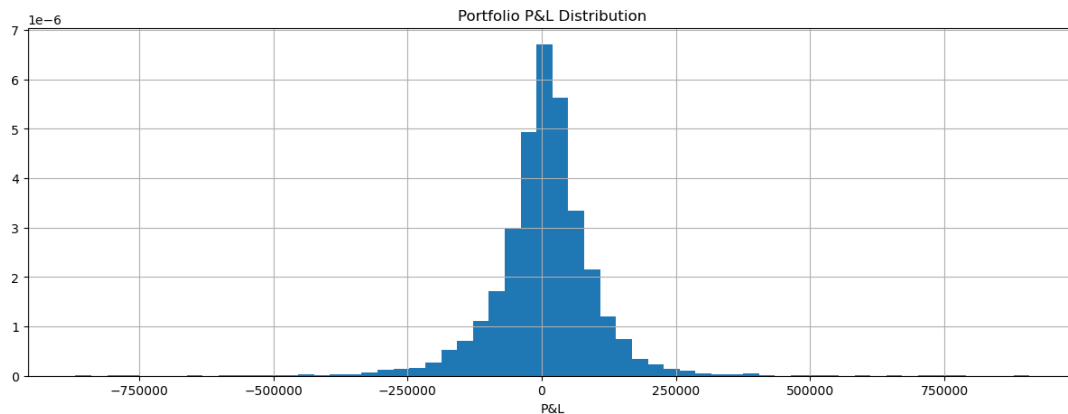


Figure 4: Empirical distribution of daily portfolio P&L.

Interpretation. This histogram is the first direct visual picture of daily risk. The distribution is centered near zero, but it is more peaked and has fatter tails than a clean bell curve. In practical terms, that means the portfolio spends most days in a fairly narrow P&L band, yet occasionally experiences much larger losses than a thin-tailed model would suggest. VaR and ES are tail summaries of this picture.

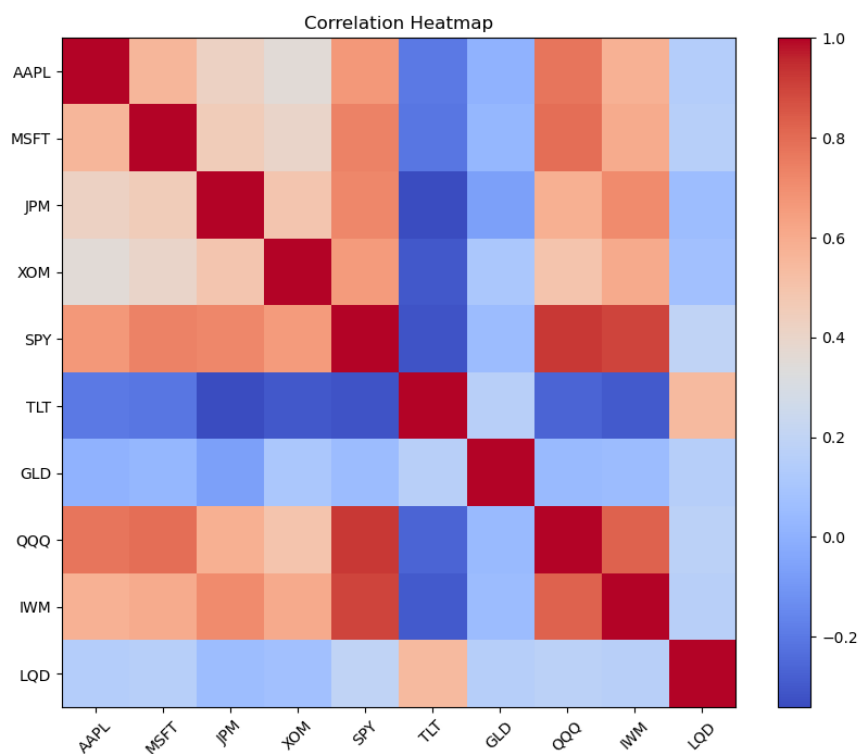


Figure 5: Unconditional correlation heatmap based on the full return sample.

Interpretation. The static correlation heatmap explains where diversification comes from. The strongest positive relationships are among the equity beta instruments, especially SPY, QQQ, and IWM. TLT appears negatively related to much of the equity sleeve, which is exactly why it can stabilise the portfolio in equity-led drawdowns. GLD sits somewhere in between: not a perfect hedge, but not simply another equity risk either. A beginner should read this chart as a map of how shocks propagate across positions.

3.2 Baseline historical simulation

Table 4: Baseline historical-simulation results.

Metric	Notebook value	Loss-magnitude interpretation
VaR 95	148,042.45	148,042.45
VaR 99	279,855.40	279,855.40
ES 95 (signed P&L)	-235,315.07	235,315.07
ES 99 (signed P&L)	-399,690.12	399,690.12

How to read the numbers. A 99% VaR of 279,855.40 means that under the plain historical-simulation model, roughly 99 days out of 100 should lose less than USD 279,855, and roughly 1 day out of 100 may lose more. The 99% ES value is shown in the notebook as signed P&L, $-399,690.12$, which corresponds to an average tail loss magnitude of USD 399,690.12 once losses have already crossed the VaR threshold.

The baseline historical-simulation results set the benchmark for everything else in the notebook. At 95%, the portfolio's VaR is about USD 148k and the corresponding tail-average loss magnitude is about USD 235k. At 99%, the threshold rises to about USD 280k while the average loss beyond the threshold rises to roughly USD 400k. This is exactly why ES matters: once the portfolio is in the worst 1% of days, losses are not merely a little worse than VaR; they are materially worse. The ratio of loss-magnitude ES to VaR is about 1.59 at 95% and about 1.43 at 99%, which is a substantial tail-severity premium.

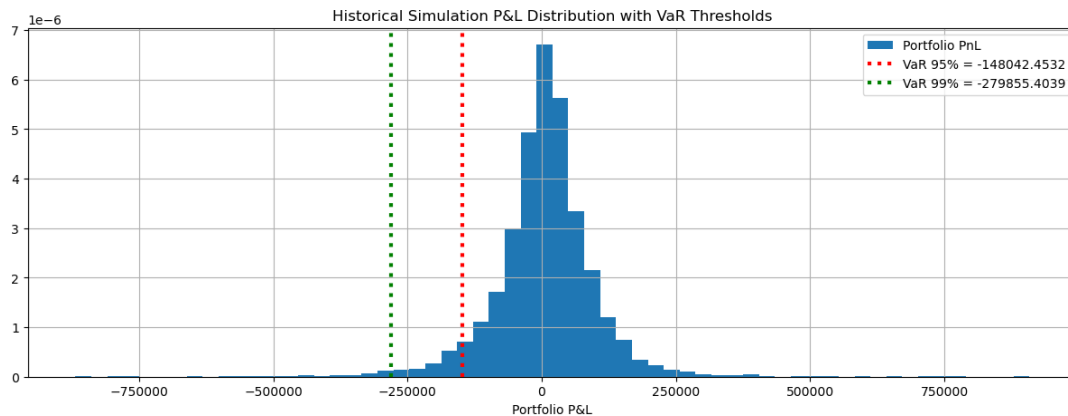


Figure 6: Historical-simulation P&L histogram with 95% and 99% VaR thresholds.

Interpretation. The vertical lines mark the cutoff between “ordinary bad days” and “statistically rare bad days” under the empirical historical distribution. Everything to the left of the red line is beyond the 95% VaR; everything to the left of the green line is beyond the 99% VaR. A good habit for a new quant is to notice that the 99% threshold sits far deeper into the left tail, and small visual changes in the far left tail can move that number a lot.

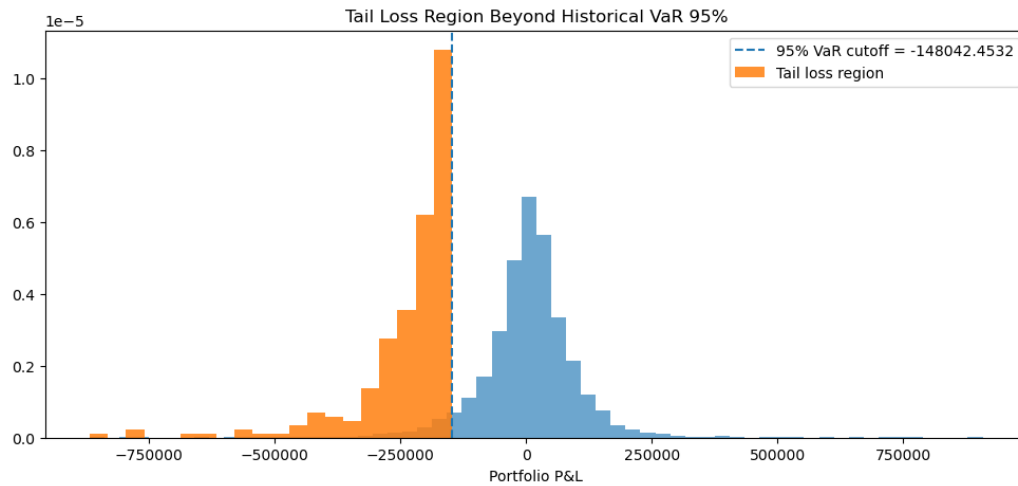


Figure 7: Tail-loss region beyond the 95% historical VaR.

Interpretation. This figure isolates the left-tail region that produces ES. The blue histogram is the overall P&L distribution; the orange overlay is the part of the distribution that lies beyond the 95% cutoff. VaR tells you where the orange region starts. ES tells you the average location *inside* that orange region. For practical risk communication, this distinction is the backbone of understanding tail risk properly.

3.3 Bootstrap historical simulation

Table 5: Bootstrap summary with sample size 100 and six bootstrap replications.

Confidence	Sample Size	Bootstraps	Mean VaR	Std VaR	VaR CI Lower	VaR CI Upper	Mean ES	Std ES	ES CI Lower	ES CI Upper
95%	100	6	138,313.30	28,974.32	105,350.21	176,357.78	-224,023.86	46,871.78	-284,824.10	-170,040.14
99%	100	6	249,130.06	54,206.68	186,730.85	324,228.10	-326,502.54	125,118.80	-510,991.38	-193,234.87

Table 6: Bootstrap summary with sample size 200 and six bootstrap replications.

Confidence	Sample Size	Bootstraps	Mean VaR	Std VaR	VaR CI Lower	VaR CI Upper	Mean ES	Std ES	ES CI Lower	ES CI Upper
95%	200	6	142,471.10	22,152.29	115,605.23	174,517.55	-237,139.35	35,858.48	-290,708.00	-200,607.76
99%	200	6	251,343.75	34,909.18	195,161.73	288,401.94	-390,116.92	112,056.08	-578,035.83	-304,021.45

The bootstrap tables quantify how unstable the HS estimates are when the sample is resampled. Because only six bootstrap replications are used in these two summary tables, the numbers should be read as a pedagogical demonstration rather than a production-grade confidence interval exercise. Even with this limitation, the message is clear. The 99% metrics have much larger standard deviations and much wider confidence intervals than the 95% metrics, because extreme-tail estimation is inherently data-hungry. Increasing the sample size from 100 to 200 generally narrows the VaR and ES uncertainty bands, especially at 95%, which is exactly what one would expect if more resampled information stabilises the estimate.

Table 7: Bootstrap convergence experiment across sample sizes and replication counts.

Bootstraps	Sample Size	Confidence	Mean VaR	Std VaR	VaR CI Width	Mean ES	Std ES	ES CI Width
6	100	95%	138,313.30	28,974.32	71,007.57	-224,023.86	46,871.78	114,783.95
6	100	99%	249,130.06	54,206.68	137,497.25	-326,502.54	125,118.80	317,756.51
6	200	95%	142,471.10	22,152.29	58,912.31	-237,139.35	35,858.48	90,100.24
6	200	99%	251,343.75	34,909.18	93,240.22	-390,116.92	112,056.08	274,014.68
12	100	95%	138,110.28	23,769.96	73,224.30	-235,439.01	50,905.03	149,232.63
12	100	99%	250,447.67	51,154.35	161,561.25	-370,386.90	142,892.66	426,808.85
12	200	95%	150,176.45	30,128.21	88,459.54	-239,540.57	41,461.82	126,729.84
12	200	99%	258,198.56	45,412.55	146,832.44	-397,751.99	128,678.39	391,628.29
24	100	95%	145,219.35	32,760.77	98,649.64	-238,800.89	52,245.27	164,742.61
24	100	99%	248,633.59	50,221.90	165,878.82	-384,669.62	162,552.59	529,171.92
24	200	95%	148,115.92	23,948.03	80,073.80	-240,862.08	36,544.36	129,668.05
24	200	99%	267,417.33	42,412.07	165,507.31	-402,090.29	113,355.23	373,486.17
36	100	95%	143,788.85	31,417.35	92,133.26	-237,245.15	52,778.19	179,979.20
36	100	99%	249,153.16	54,604.65	185,412.01	-380,328.40	161,324.02	499,921.14
36	200	95%	147,215.89	26,366.64	86,656.21	-238,490.13	46,754.75	162,743.57
36	200	99%	274,149.94	68,723.25	225,363.32	-393,617.29	120,392.51	434,205.92

The convergence experiment broadens the same lesson. As the number of bootstrap replications rises from 6 to 36, the mean estimates move around less erratically, but the 99% tail metrics still remain visibly unstable. For example, the 99% mean VaR for sample size 200 ranges from about USD 251k to USD 274k across the different bootstrap counts in the table. That is a large spread for a supposedly “single” risk number, and it teaches an important operational lesson: very-high-confidence tail measures should never be quoted without acknowledging estimation uncertainty.

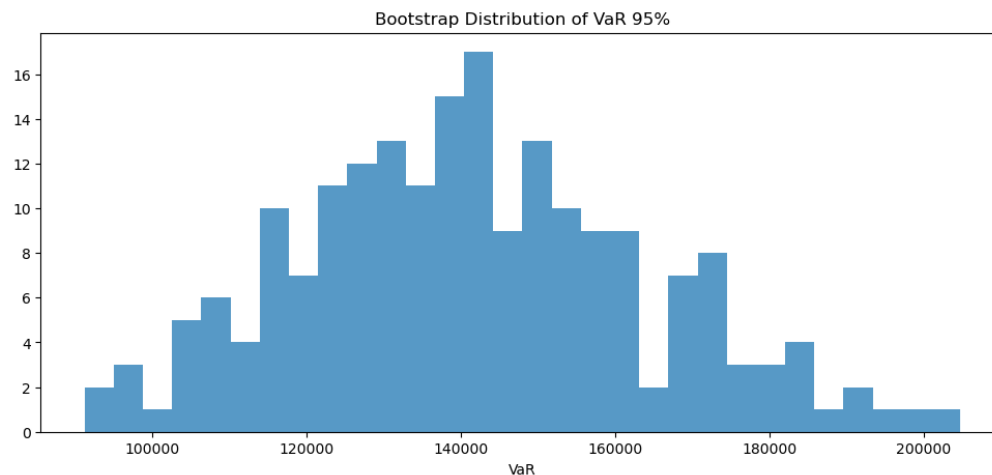


Figure 8: Bootstrap distribution of the 95% VaR estimate.

Interpretation. The 95% bootstrap histogram is fairly concentrated around the mid-USD 100k range. That means the 95% VaR estimate is comparatively stable. There is still noticeable sampling variability, but the mass is compact enough that the analyst can have moderate confidence in the order of magnitude of the estimate.

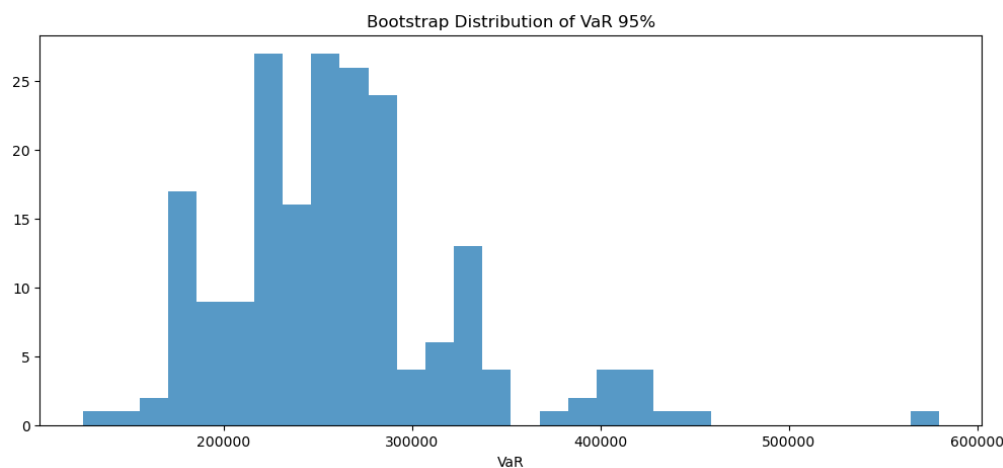


Figure 9: Bootstrap distribution of the 99% VaR estimate.

Interpretation. The 99% bootstrap histogram is wider and visibly more irregular. That is exactly what practitioners mean when they say that extreme quantiles are noisy. The tail is being estimated from very few truly extreme observations, so resampling changes the answer more dramatically.

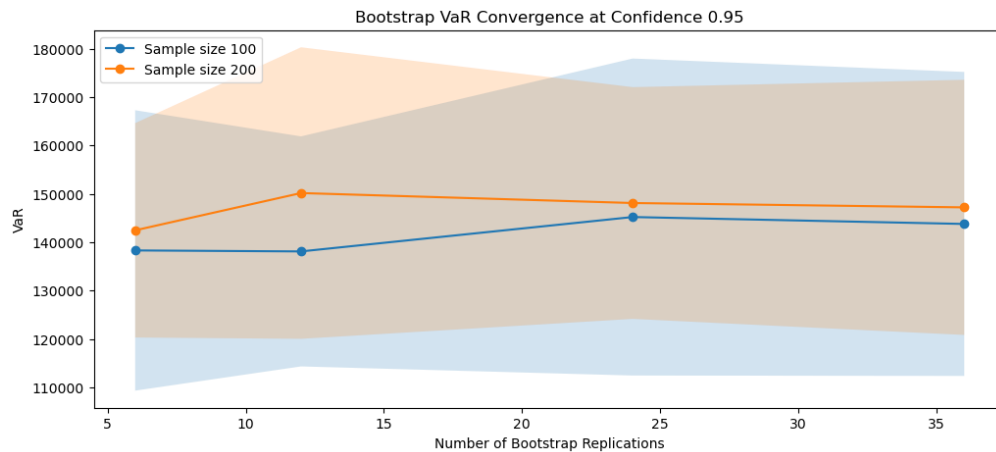


Figure 10: Bootstrap VaR convergence at 95% confidence for sample sizes 100 and 200.

Interpretation. The 95% convergence plot shows the mean bootstrap VaR and one-standard-deviation bands as the number of replications increases. The sample-size-200 line is typically more stable than the sample-size-100 line. In practical terms, a larger effective sample makes the estimate less sensitive to resampling noise.

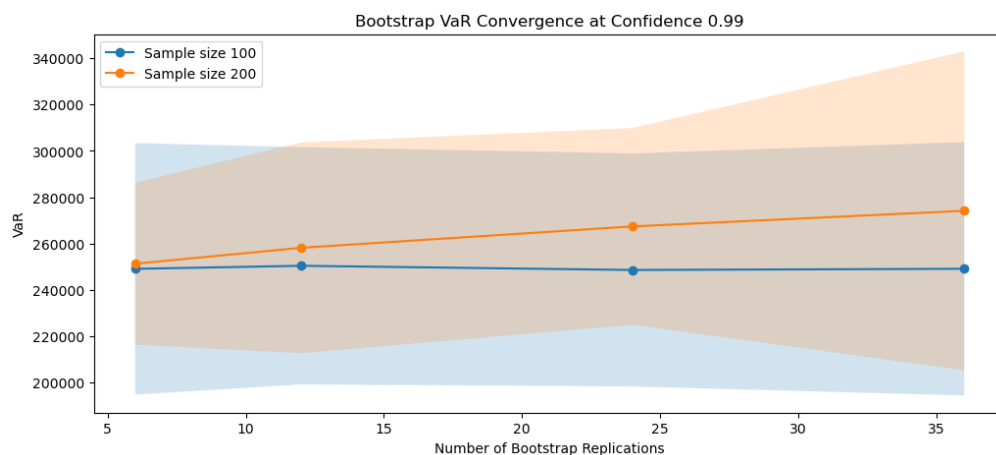


Figure 11: Bootstrap VaR convergence at 99% confidence for sample sizes 100 and 200.

Interpretation. The 99% convergence plot is the more important training figure. Even after increasing the replication count, the uncertainty bands stay wide. This is a strong reminder that an analyst should never overstate the precision of a 99% VaR number derived from one historical sample.

3.4 Exponentially weighted historical simulation

Table 8: Exponentially weighted historical-simulation output. The third column converts the raw notebook output into a consistent positive loss convention.

Metric	Raw notebook output	Consistent loss view
VaR 95%	-124,576.36	124,576.36
ES 95%	155,644.38	155,644.38
VaR 99%	-180,145.10	180,145.10
ES 99%	179,537.05	179,537.05

The EWHS output is numerically smaller than the baseline HS output, but the first thing to notice is not the level of risk; it is the sign inconsistency. The notebook prints VaR as a negative P&L quantile while it prints ES as a positive loss magnitude. For interpretation, the meaningful comparison is therefore the third column. On that consistent basis, the EWHS 95% loss threshold is about USD 124.6k and the 99% loss threshold is about USD 180.1k, both materially below plain HS.

Practical meaning of the EWHS result. Because $\hat{\lambda} \approx 0.9419$, the weighting scheme has a half-life of only about 11.6 days. That is an extremely short memory relative to a sample running back to 2007. So the model is effectively saying: “Recent conditions matter overwhelmingly more than old crises.” That choice naturally pulls VaR down when the most recent regime is calmer than 2008 or 2020.

Table 9: Top 20 worst historical portfolio P&L observations and their exponentially weighted probabilities.

Date	pnl	exp_weight	cum_weight
2020-03-16	-868,202.49	5.828e-41	5.828e-41
2008-09-29	-785,093.61	6.831e-116	5.828e-41
2020-03-12	-776,272.83	5.171e-41	1.100e-40
2008-12-01	-654,671.20	9.499e-115	1.100e-40
2020-03-09	-650,273.44	4.321e-41	1.532e-40
2008-10-15	-579,783.55	1.400e-115	1.532e-40
2020-03-18	-549,190.38	6.569e-41	2.189e-40
2009-01-20	-526,479.36	6.840e-114	2.189e-40
2008-11-20	-502,703.62	6.634e-115	2.189e-40
2020-06-11	-459,468.16	2.241e-39	2.460e-39
2008-11-19	-449,399.49	6.249e-115	2.460e-39
2025-04-04	-439,175.74	6.140e-08	6.140e-08
2011-08-08	-433,821.57	3.477e-97	6.140e-08
2008-10-09	-431,640.07	1.102e-115	6.140e-08
2008-10-07	-429,671.38	9.781e-116	6.140e-08
2025-04-03	-415,739.69	5.784e-08	1.192e-07
2008-11-05	-402,575.49	3.435e-115	1.192e-07
2020-03-11	-400,343.10	4.870e-41	1.192e-07
2008-09-17	-386,575.29	4.233e-116	1.192e-07
2008-11-14	-372,046.96	5.222e-115	1.192e-07

The top-20-loss table makes the recency effect concrete. Some of the worst losses in the entire history occurred during the Global Financial Crisis and the COVID shock, yet their exponential weights are essentially zero. By contrast, losses from April 2025 receive visibly non-negligible weight because they are recent enough to matter under such a short half-life. This table is one of the best illustrations in the notebook of how model design choices directly change economic interpretation.

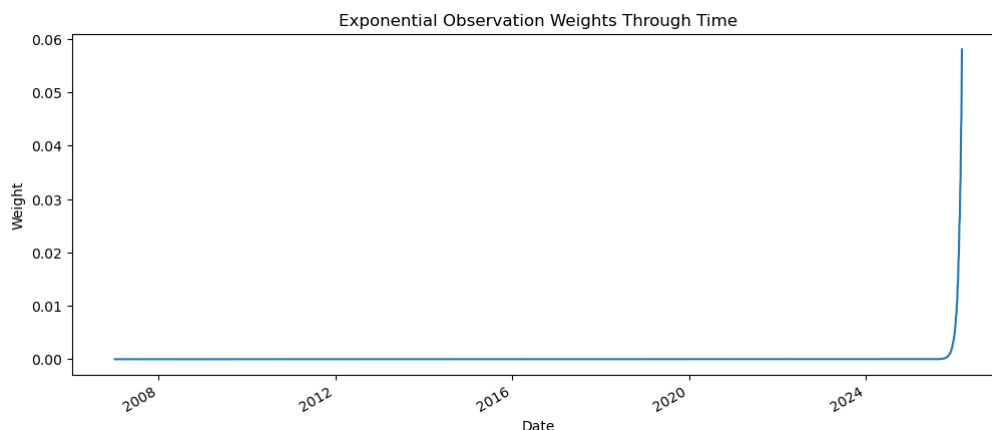


Figure 12: Exponential observation weights through time.

Interpretation. This plot shows the weight schedule directly. Almost the entire probability mass is piled up at the far right edge of the sample. For a beginner, this is the picture that turns an abstract λ parameter into intuition: the method is not “historical” in the equal-weight sense any more; it is a heavily recent-history model.

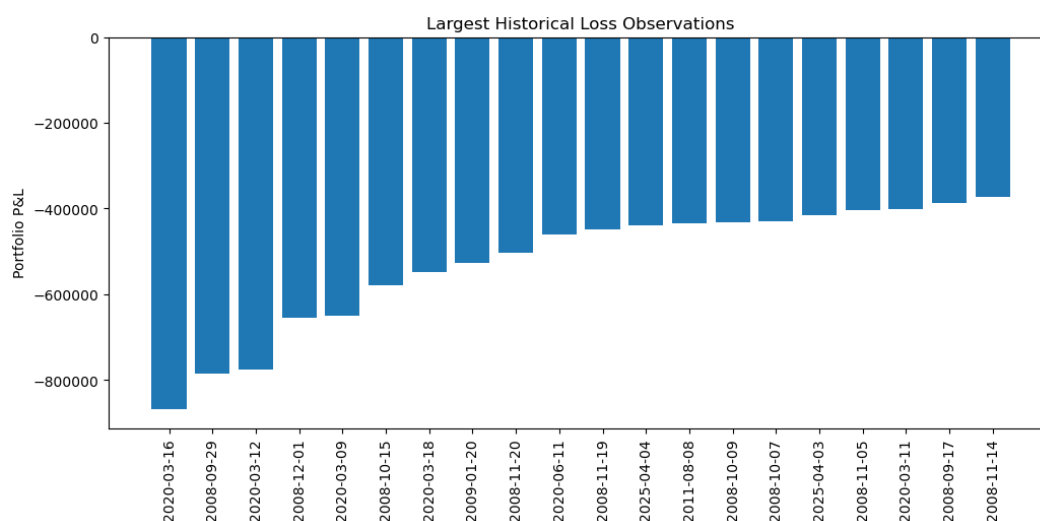


Figure 13: Largest historical portfolio loss observations.

Interpretation. The worst losses come from familiar stress windows, especially March 2020 and late 2008. This is important because plain HS treats all of them as equally relevant pieces of the empirical tail, whereas EWSH does not.

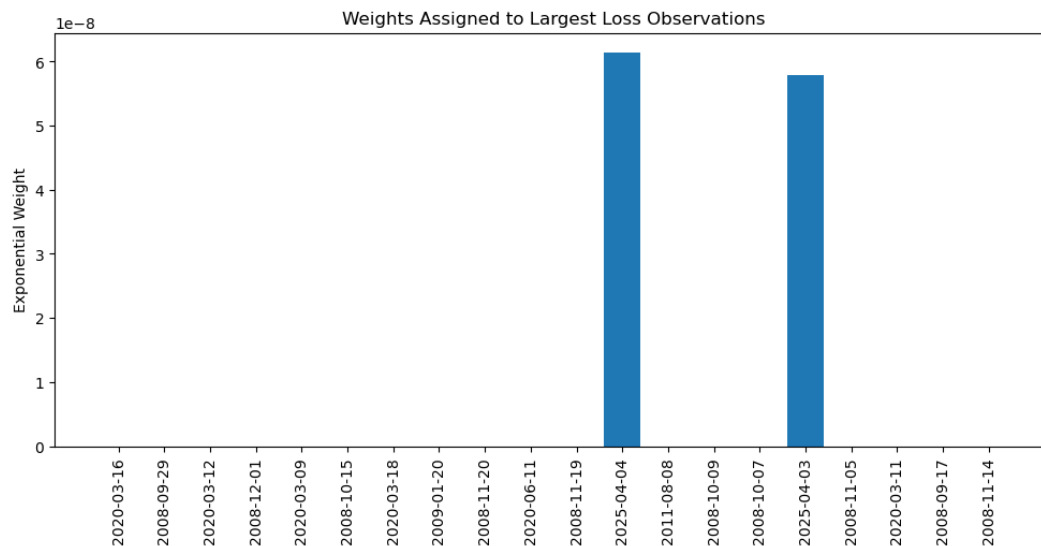


Figure 14: Exponential weights assigned to the largest historical losses.

Interpretation. Here the contrast becomes explicit. Most of the catastrophic legacy observations receive almost no weight. Only the more recent large losses carry visible probability mass. An analyst should read this figure and immediately understand why the EWHS VaR is lower than the plain HS VaR.

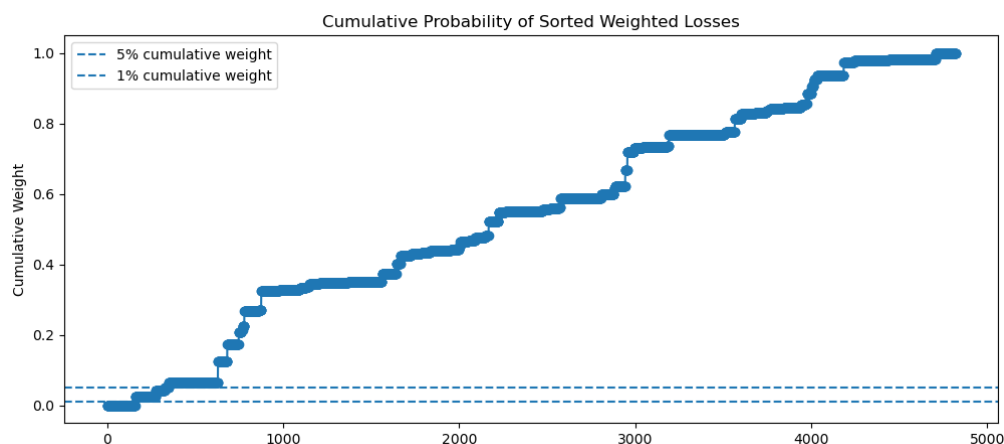


Figure 15: Cumulative probability of sorted weighted losses.

Interpretation. The dashed horizontal lines show the 5% and 1% cumulative-probability levels used for 95% and 99% VaR. Because the weights are so concentrated, the weighted quantile is determined by a very small recent subset of observations rather than by the full historical tail. This is a powerful teaching figure: the same return history can imply very different VaR numbers depending on the probability measure imposed on past observations.

3.5 Volatility-weighted historical simulation

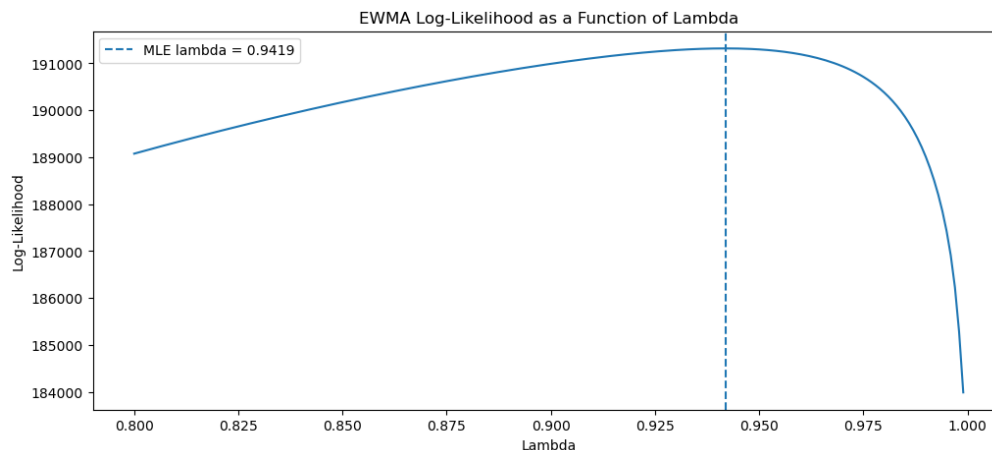


Figure 16: EWMA log-likelihood as a function of lambda.

Interpretation. The likelihood curve peaks around $\lambda \approx 0.9419$. Very low λ would make volatility react too fast and become noisy; very high λ would make it too sluggish. The estimated optimum therefore balances responsiveness and smoothness. In practical risk work, this plot is the diagnostic that says the decay parameter was not chosen arbitrarily.

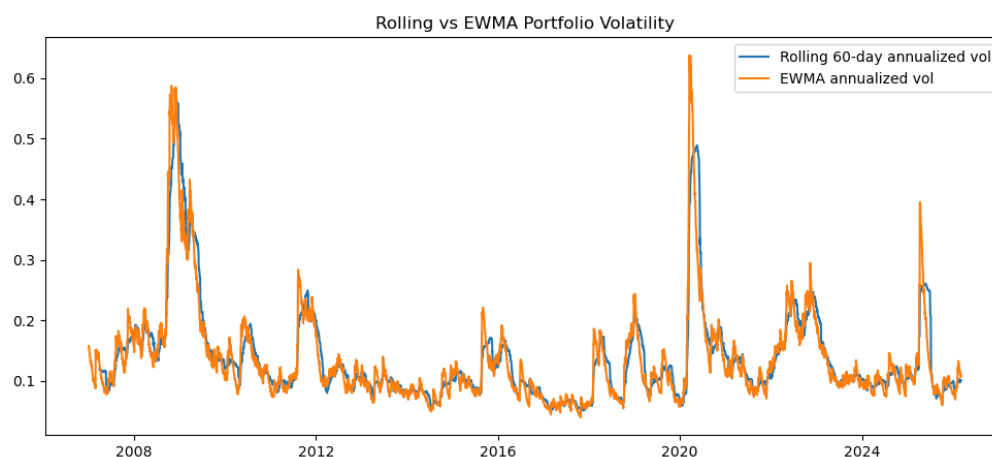


Figure 17: Rolling 60-day annualised volatility versus EWMA annualised volatility.

Interpretation. The rolling and EWMA volatility measures broadly agree on the major stress episodes, but the EWMA series reacts faster and more smoothly. That is exactly what filtered historical simulation is meant to exploit. It uses a volatility estimate that adapts in real time and then re-expresses historical shocks under the current volatility regime.

Table 10: Volatility-weighted historical-simulation results.

Metric	Notebook value	Loss-magnitude interpretation
VaR 95%	138,169.98	138,169.98
VaR 99%	235,104.28	235,104.28
ES 95% (signed P&L)	-199,619.35	199,619.35
ES 99% (signed P&L)	-303,110.73	303,110.73

Relative to plain HS, the volatility-weighted results are somewhat lower: the 95% VaR falls from about USD 148k to USD 138k, and the 99% VaR falls from about USD 280k to USD 235k. The corresponding ES magnitudes also fall. The interpretation is that once old shocks are normalised for today's volatility regime, the tail looks less severe than in the unconditional historical sample. This is a sensible result if the current volatility regime is calmer than the crisis episodes embedded in the full history.

3.6 Correlation-weighted historical simulation

Table 11: Current EWMA covariance matrix printed by the notebook.

Ticker	AAPL	MSFT	JPM	XOM	SPY	TLT	GLD	QQQ	IWM	LQD
AAPL	0.000272	0.000025	0.000115	-0.000017	0.000065	-0.000023	-0.000051	0.000068	0.000084	-0.000004
MSFT	0.000025	0.000422	0.000091	-0.000081	0.000057	-0.000029	-0.000027	0.000086	0.000060	-0.000010
JPM	0.000115	0.000091	0.000285	-0.000048	0.000077	-0.000008	-0.000010	0.000076	0.000136	0.000004
XOM	-0.000017	-0.000081	-0.000048	0.000280	-0.000007	-0.000010	0.000196	-0.000017	-0.000006	-0.000007
SPY	0.000065	0.000057	0.000077	-0.000007	0.000060	-0.000004	0.000029	0.000078	0.000085	0.000003
TLT	-0.000023	-0.000029	-0.000008	-0.000010	-0.000004	0.000028	-0.000007	-0.000008	-0.000004	0.000010
GLD	-0.000051	-0.000027	-0.000010	0.000196	0.000029	-0.000007	0.000634	0.000051	0.000091	-0.000003
QQQ	0.000068	0.000086	0.000076	-0.000017	0.000078	-0.000008	0.000051	0.000113	0.000102	0.000003
IWM	0.000084	0.000060	0.000136	-0.000006	0.000085	-0.000004	0.000091	0.000102	0.000177	0.000008
LQD	-0.000004	-0.000010	0.000004	-0.000007	0.000003	0.000010	-0.000003	0.000003	0.000008	0.000006

Table 12: Current EWMA correlation matrix printed by the notebook.

Ticker	AAPL	MSFT	JPM	XOM	SPY	TLT	GLD	QQQ	IWM	LQD
AAPL	1.000	0.073	0.413	-0.061	0.506	-0.266	-0.122	0.385	0.381	-0.100
MSFT	0.073	1.000	0.263	-0.236	0.360	-0.265	-0.051	0.395	0.221	-0.209
JPM	0.413	0.263	1.000	-0.170	0.593	-0.090	-0.023	0.425	0.606	0.092
XOM	-0.061	-0.236	-0.170	1.000	-0.056	-0.107	0.465	-0.098	-0.027	-0.179
SPY	0.506	0.360	0.593	-0.056	1.000	-0.102	0.150	0.948	0.830	0.172
TLT	-0.266	-0.265	-0.090	-0.107	-0.102	1.000	-0.056	-0.149	-0.050	0.826
GLD	-0.122	-0.051	-0.023	0.465	0.150	-0.056	1.000	0.190	0.272	-0.050
QQQ	0.385	0.395	0.425	-0.098	0.948	-0.149	0.190	1.000	0.721	0.102
IWM	0.381	0.221	0.606	-0.027	0.830	-0.050	0.272	0.721	1.000	0.262
LQD	-0.100	-0.209	0.092	-0.179	0.172	0.826	-0.050	0.102	0.262	1.000

The covariance and correlation matrices show the current dependence structure implied by the EWMA filter. Several relationships are economically intuitive and very important for portfolio risk: SPY–QQQ is extremely high at about 0.948, SPY–IWM is also high at about 0.830, and TLT–LQD is high at about 0.826. Those numbers tell us that the portfolio’s apparent diversification is not as strong as the asset count alone might suggest, because much of the equity sleeve loads on the same broad market factor.

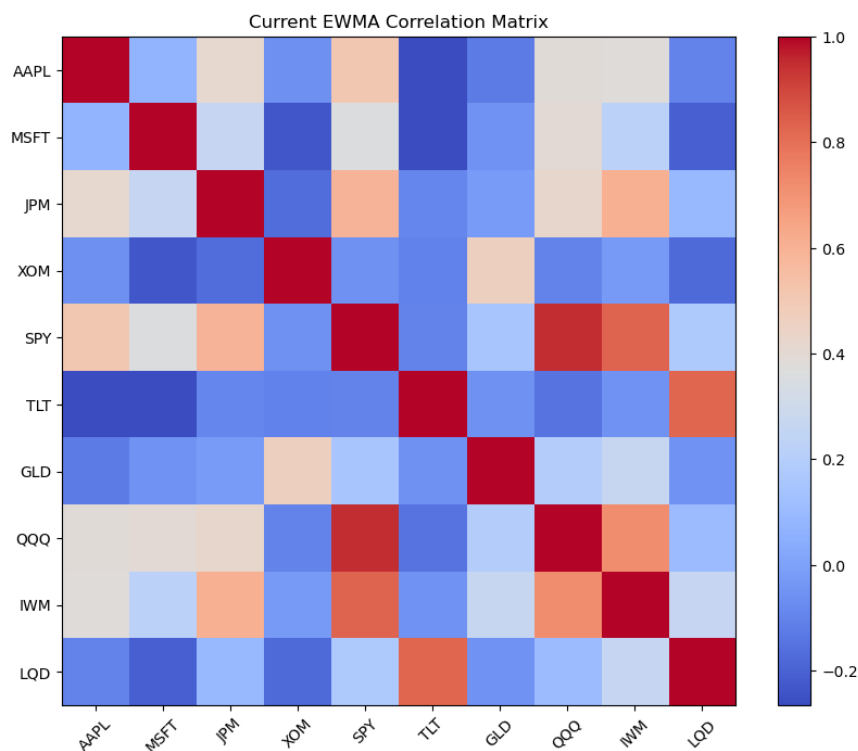


Figure 18: Current EWMA correlation heatmap.

Interpretation. Compared with the unconditional heatmap earlier, this figure is about *current* dependence, not average dependence over the full history. That distinction matters enormously in risk management. Diversification lives or dies on current correlation, not historical average correlation.

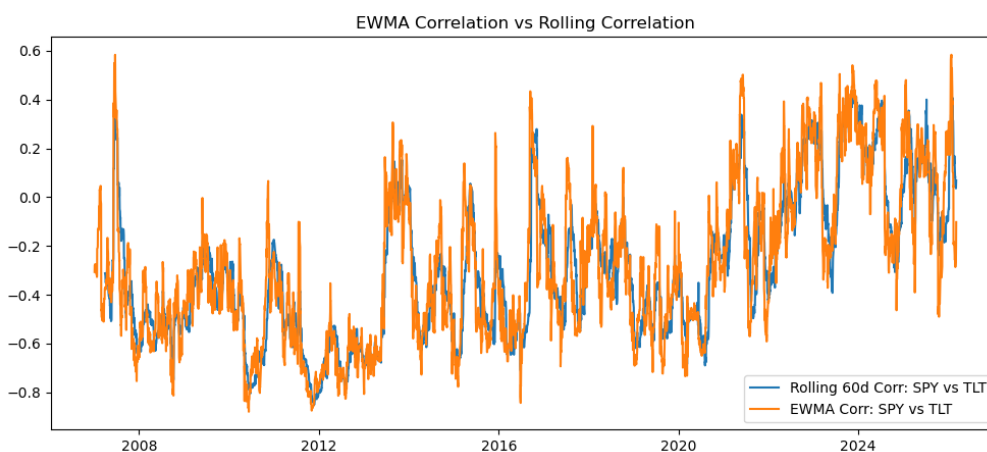


Figure 19: EWMA correlation versus rolling 60-day correlation for SPY and TLT.

Interpretation. The SPY–TLT relationship is not constant. Sometimes it is strongly negative, offering diversification, and sometimes it is much weaker. The EWMA estimate moves with the regime while remaining smoother than the rolling-window estimate. This is exactly why correlation-adjusted risk models can produce different answers from unconditional HS.

Table 13: Comparison of baseline, volatility-weighted, and correlation-weighted historical simulation.

Model	VaR 95%	VaR 99%	ES 95%	ES 99%
Historical Simulation	148,042.45	279,855.40	-235,315.07	-399,690.12
Volatility Weighted HS	138,169.98	235,104.28	-199,619.35	-303,110.73
Correlation Weighted HS	104,474.22	150,497.90	-134,627.67	-176,196.13

The model-comparison table shows a clear ranking. Relative to baseline HS, volatility-weighted HS lowers the 95% VaR by about 6.7% and the 99% VaR by about 16.0%. Correlation-weighted HS lowers them even more sharply—about 29.4% at 95% and 46.2% at 99%. Economically, that means the current covariance regime implied by the filter is considerably milder than the full-sample crisis regime embedded in plain HS.

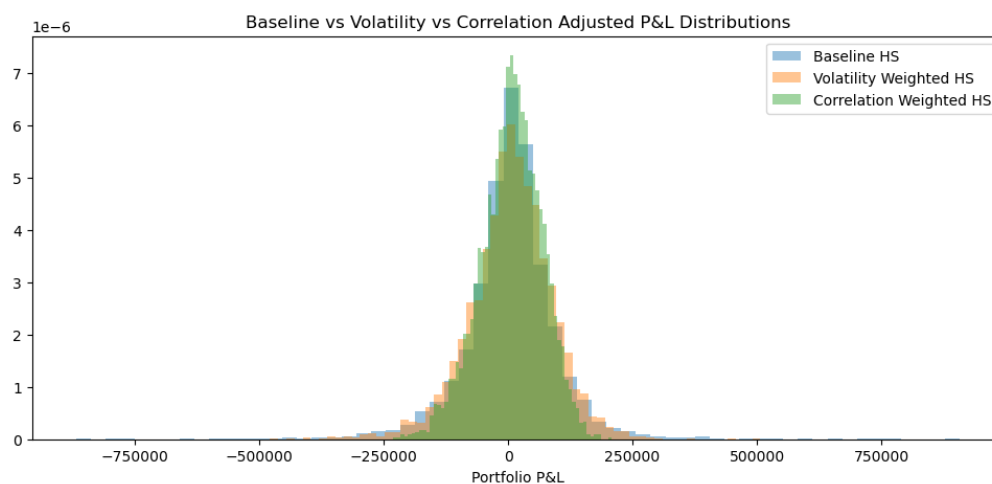


Figure 20: Overlay of baseline, volatility-weighted, and correlation-weighted P&L distributions.

Interpretation. The overlay makes the comparison visual. The correlation-adjusted distribution is the tightest, which is why it produces the smallest VaR and ES. For a new analyst, the key lesson is not just “different models give different numbers.” It is *why* they do: the models are using different assumptions about which historical states are relevant and how those states should be translated into today’s market regime.

3.7 VaR backtesting

Table 14: One-day 99% VaR backtesting summary from the notebook.

Model	Failures	Failure Rate	Expected Failures	Kupiec LR	Kupiec p-value	Christoffersen LR_ind	Christoffersen p_ind	Christo
Historical Simulation	50	1.16%	43.23	1.020	0.3126	23.819	1.00e-06	24.839
Exponentially Weighted HS	4230	97.85%	43.23	38063.551	0.00e+00	1.632	0.2014	38065.1
Volatility Weighted HS	50	1.16%	43.23	1.020	0.3126	13.546	0.0002	14.566
Correlation Weighted HS	111	2.57%	43.23	74.884	0.00e+00	6.716	0.0096	81.600
Delta-Normal VaR	84	1.94%	43.23	30.449	3.43e-08	21.291	4.00e-06	51.740

This is one of the most valuable tables in the notebook because it separates in-sample attractiveness from out-of-sample credibility.

- **Historical Simulation.** It records 50 breaches against an expected 43.23, so the frequency is close enough that Kupiec does not reject the model ($p \approx 0.313$). However, Christoffersen strongly rejects independence ($p_{cc} \approx 4.04 \times 10^{-6}$). That means the model gets the average count roughly right but the breaches arrive in clusters.
- **Exponentially Weighted HS.** The notebook reports 4,230 failures out of roughly 4,323 forecasts, which is economically impossible for a valid 99% VaR model. This is not evidence that EWHS is catastrophically bad; it is evidence of the earlier sign-convention bug. Because the EWHS forecast VaR is negative in the implementation, the breach rule compares realized P&L to a positive threshold and almost every day is counted as a breach.
- **Volatility-Weighted HS.** It has the same number of failures as plain HS and the same Kupiec p-value, so the unconditional hit rate is similarly acceptable. But independence still fails, though somewhat less severely than under plain HS. This is a common real-world pattern: filtering helps, but not enough to eliminate crisis clustering.
- **Correlation-Weighted HS.** It records 111 breaches, far above the expected 43.23, and therefore fails both unconditional and conditional coverage tests. This means the model is too optimistic out of sample; its VaR threshold is too small.
- **Delta-Normal VaR.** It also has too many breaches (84) and fails both tests, which is consistent with the general concern that Gaussian assumptions understate tail risk in crisis-prone markets.

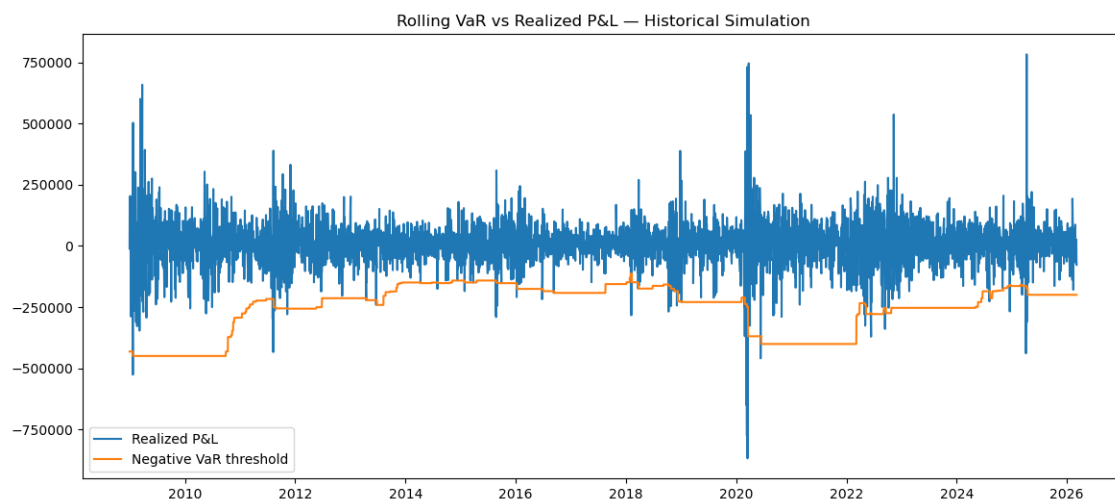


Figure 21: Rolling 99% historical-simulation VaR versus realized P&L.

Interpretation. When the realized P&L line drops below the negative VaR threshold line, a breach occurs. Notice that breaches do not appear uniformly over time. They bunch up in turbulent regimes. This is the visual explanation for why unconditional coverage can look okay while conditional coverage fails.

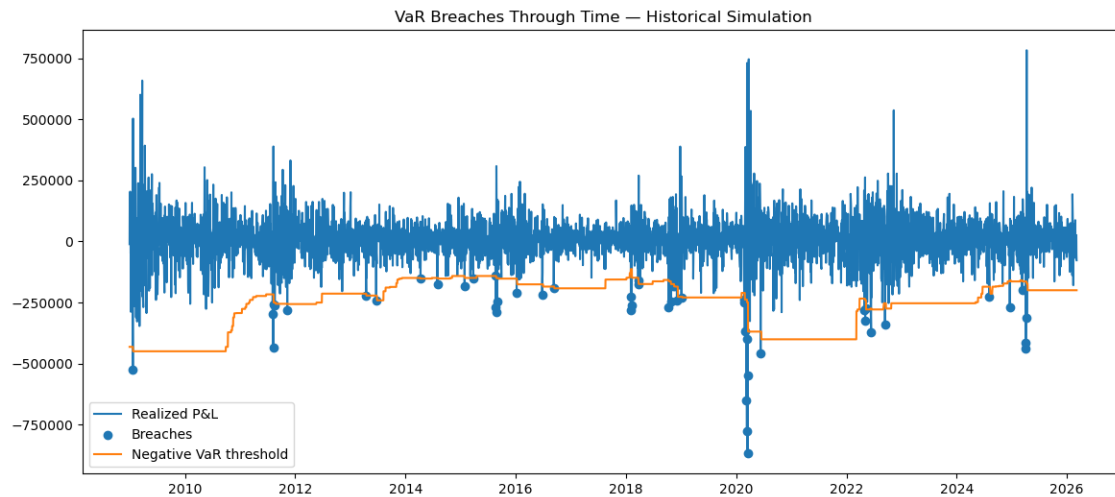


Figure 22: Historical-simulation VaR breaches highlighted through time.

Interpretation. The scatter points make the clustering unmistakable. In risk management practice, clustered exceptions are a warning sign that market conditions are shifting faster than the model's effective assumptions. It is one of the cleanest practical demonstrations in the notebook of why backtesting should never stop at raw exception counts.

3.8 Scenario analysis

Table 15: Scenario analysis from the notebook.

Scenario	Portfolio Return	Portfolio P&L
Equity Crash -10%	-6.98%	-698,000.00
Rates Shock	-3.64%	-364,000.00
Commodity Shock	-2.30%	-230,000.00
Volatility Spike	-5.16%	-516,000.00

The scenario table is easy to read and extremely useful for communication. An equity crash of -10% across the equity sleeves produces a portfolio loss of about USD 698k, by far the worst of the four scenarios. That is economically consistent with the portfolio composition: most of the weight sits in equity-like exposures. The volatility-spike scenario is next at roughly USD 516k, followed by rates shock and commodity shock.

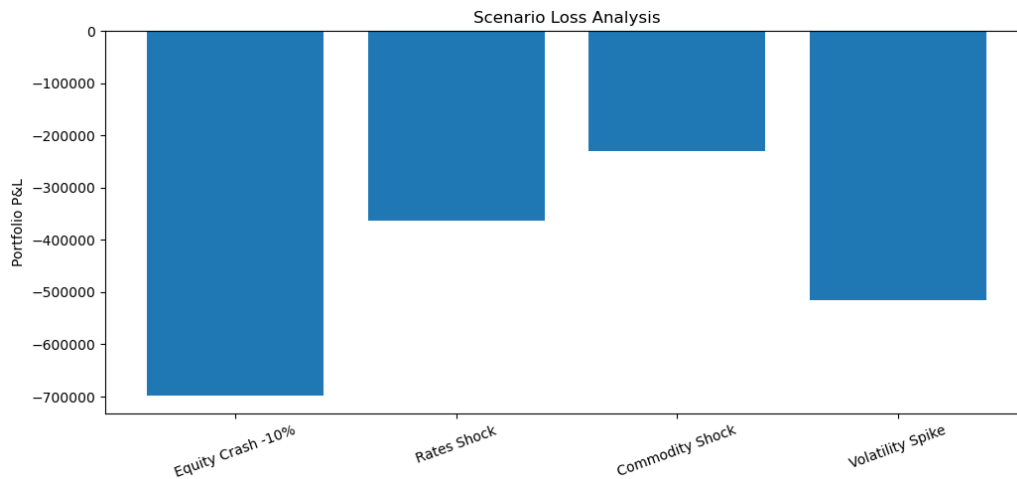


Figure 23: Scenario loss analysis.

Interpretation. This bar chart translates model risk into intuitive business language. The largest scenario loss is around 2.5 times the plain 99% historical VaR, which means a plausible designed stress can be much worse than the “1 day in 100” historical threshold. This is exactly why VaR should not be the only number seen by senior management.

3.9 Historical stress testing

Table 16: Historical stress-test results from the notebook.

Stress Window	Start	End	Worst Day	VaR 95%	VaR 99%	ES 97.5%
GFC	2008-09-01	2009-03-31	-785,093.61	422,897.41	620,971.76	-636,506.93
COVID Shock	2020-02-20	2020-04-30	-868,202.49	604,786.06	823,156.96	-822,237.66
2022 Inflation Shock	2022-01-01	2022-10-31	-371,105.59	204,336.19	322,173.77	-307,416.88

The historical stress table re-runs tail metrics inside specific crisis windows. The COVID shock window is the most severe of the three: worst daily loss about USD 868k, 95% VaR about USD 605k, and 99% VaR about USD 823k. The Global Financial Crisis is the next most severe, while the 2022 inflation shock is materially milder for this portfolio.

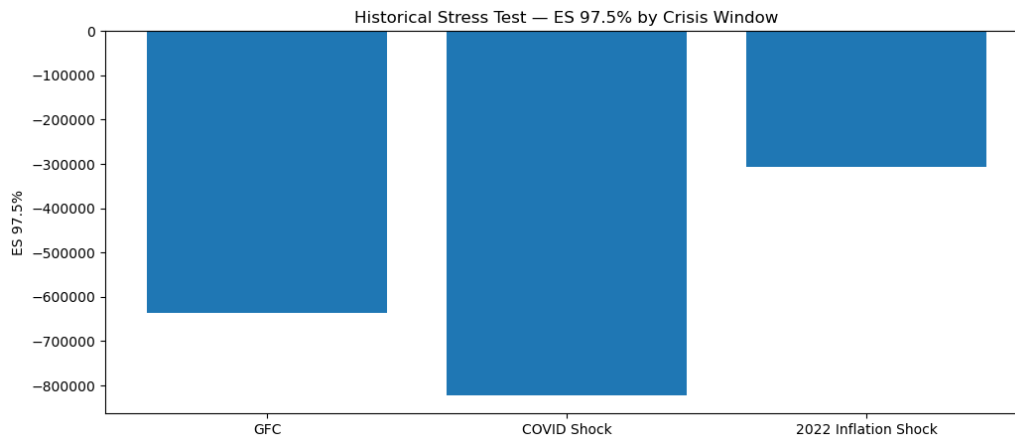


Figure 24: Historical stress-window ES at 97.5%.

Interpretation. The bar chart shows that crisis-window ES is far larger in magnitude than ordinary full-sample daily VaR. The COVID-window 99% VaR is almost three times the baseline full-sample 99% VaR. That gap is the practical definition of regime dependence. It tells the analyst that a calm unconditional daily VaR number can seriously understate what happens when the market enters a true stress regime.

4 Summary

Table 17: Selected headline risk numbers in positive loss language.

Measure	Value
Baseline HS VaR 99%	279,855.40
Baseline HS ES 99% (loss magnitude)	399,690.12
Volatility-Weighted HS VaR 99%	235,104.28
Correlation-Weighted HS VaR 99%	150,497.90
Equity crash scenario loss	698,000.00
COVID stress-window VaR 99%	823,156.96
Worst historical daily loss in COVID window	868,202.49

This notebook provides an excellent compact tour of modern daily market-risk measurement. The core lessons are the following.

1. **Baseline historical simulation gives a sensible starting point.** For this USD 10 million portfolio, the plain historical 99% VaR is about USD 280k and the corresponding average tail loss magnitude is about USD 400k. Those are the notebook's benchmark one-day numbers.
2. **Tail severity is materially worse than the threshold itself.** The gap between VaR and ES is large enough that any serious discussion of tail risk should include both.
3. **Bootstrap results show that extreme-tail estimation is noisy.** The 99% tail is much less stable than the 95% tail, especially when bootstrap counts are small. That is a statistical uncertainty point, not a market-view point.

4. **Recency-weighted models can dramatically shrink measured risk.** The estimated EWMA decay implies very short memory, so old crisis observations receive almost no weight. That explains why EWHS and filtered historical methods can be materially below plain HS.
5. **Backtesting matters more than in-sample neatness.** Plain HS and volatility-weighted HS get the breach count roughly right, but both still suffer from breach clustering. Correlation-weighted HS and delta-normal VaR underestimate tail risk out of sample. The EWHS backtest in the notebook is contaminated by a sign bug and should not be interpreted as a true model failure.
6. **Stress analysis is indispensable.** Scenario losses and crisis-window stress metrics are far larger than ordinary daily VaR numbers. In practice, this means VaR is a useful day-to-day threshold metric, but it is not a complete description of market risk.

Bottom line for a new quant risk analyst. Read the portfolio distribution first. Then read VaR as the tail threshold and ES as the average loss beyond that threshold. After that, ask how the model treats old crises, current volatility, and current correlation. Finally, do not trust any VaR number until you have checked backtests and stress tests. That sequence captures the practical workflow of interpreting VaR correctly.

5 Appendix: derivations

5.1 From price to P&L

Starting from adjusted prices $P_{i,t}$, the notebook computes

$$r_{i,t} = \ln(P_{i,t}) - \ln(P_{i,t-1}), \quad R_{i,t} = e^{r_{i,t}} - 1.$$

For weights that sum to one,

$$R_{p,t} = \sum_{i=1}^n w_i R_{i,t}, \quad \Pi_t = V_0 R_{p,t}.$$

Hence the portfolio P&L distribution is just the weighted asset-return distribution scaled by portfolio value.

5.2 Historical-simulation VaR and ES

Let $\Pi_{(1)} \leq \Pi_{(2)} \leq \dots \leq \Pi_{(T)}$ denote the ordered P&L sample. If $k = \lceil \alpha T \rceil$, then a discrete historical VaR estimator is

$$\widehat{\text{VaR}}_c^{HS} = -\Pi_{(k)}.$$

The corresponding signed-P&L ES estimator is

$$\widehat{\text{ES}}_c^{HS} = \frac{1}{k} \sum_{j=1}^k \Pi_{(j)}.$$

If one prefers positive loss language, multiply by -1 .

5.3 Weighted quantile and weighted ES

For weighted historical simulation, let sorted observations be $x_{(1)} \leq \dots \leq x_{(T)}$ with associated non-negative weights $w_{(1)}, \dots, w_{(T)}$ summing to one. The weighted α -quantile is the smallest point where the cumulative weight reaches α :

$$Q_\alpha^w = \inf \left\{ x : \sum_{j: x_{(j)} \leq x} w_{(j)} \geq \alpha \right\}.$$

A weighted ES is then the weighted mean of all observations up to that cutoff, with a fractional final observation if interpolation is used. That is exactly the logic implemented in the notebook's weighted ES function.

5.4 EWMA likelihood and lambda estimation

Suppose for a centered return series u_t that

$$u_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2), \quad \sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) u_{t-1}^2.$$

Ignoring constants, the Gaussian negative log-likelihood over one series is

$$\ell(\lambda) = \frac{1}{2} \sum_{t=2}^T \left(\ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2} \right).$$

The notebook sums this objective over all assets and numerically minimises it over $\lambda \in [0.80, 0.999]$. The maximiser determines how quickly conditional variance is allowed to adapt.

5.5 Volatility scaling derivation

If a historical return r_t occurred when conditional volatility was σ_t , but today's volatility is σ_T , a natural filtered-historical adjustment is

$$\tilde{r}_t = \mu + (r_t - \mu) \frac{\sigma_T}{\sigma_t}.$$

This preserves the return's sign and standardized size $(r_t - \mu)/\sigma_t$, while rewriting it under today's volatility level. Applying this componentwise across assets yields the filtered return matrix used in volatility-weighted HS.

5.6 Correlation adjustment derivation

Let $\mathbf{S}_t = \mathbf{L}_t \mathbf{L}_t^\top$ be the historical covariance matrix on date t , and let $\mathbf{S}_T = \mathbf{L}_T \mathbf{L}_T^\top$ be the current covariance matrix. For a historical mean-adjusted return vector $\mathbf{r}_t - \boldsymbol{\mu}$, define the standardized shock

$$\mathbf{z}_t = \mathbf{L}_t^{-1} (\mathbf{r}_t - \boldsymbol{\mu}).$$

Then $\mathbf{z}_t$ is the historical shock expressed in “unit covariance” coordinates. Re-imposing today's covariance gives

$$\tilde{\mathbf{r}}_t = \boldsymbol{\mu} + \mathbf{L}_T \mathbf{z}_t.$$

So the model keeps the historical shock direction but expresses it in the current dependence regime.

5.7 Delta-normal VaR and ES derivation

Assume portfolio return $R_p \sim N(\mu_p, \sigma_p^2)$. Write $z_c = \Phi^{-1}(c)$ and $\alpha = 1 - c$. The α -quantile of R_p is

$$q_\alpha = \mu_p - z_c \sigma_p.$$

Therefore portfolio-loss VaR in dollars is

$$\text{VaR}_c^{DN} = -V_0 q_\alpha = V_0(z_c \sigma_p - \mu_p).$$

For ES, use the truncated-normal identity

$$\mathbb{E}[Z \mid Z \leq -z_c] = -\frac{\phi(z_c)}{\alpha}, \quad Z \sim N(0, 1).$$

Then

$$\mathbb{E}[R_p \mid R_p \leq q_\alpha] = \mu_p - \sigma_p \frac{\phi(z_c)}{\alpha}.$$

Negating this signed tail mean yields the positive-loss ES formula

$$\text{ES}_c^{DN} = V_0 \left(\frac{\phi(z_c)}{1-c} \sigma_p - \mu_p \right).$$

5.8 Kupiec test derivation

Let I_t be the VaR-breach indicator and $x = \sum_{t=1}^T I_t$. Under a correctly calibrated VaR model with tail probability α , the likelihood under the null is

$$L_0 = (1 - \alpha)^{T-x} \alpha^x.$$

Under the alternative, the breach probability is estimated by $\hat{\pi} = x/T$, giving

$$L_1 = (1 - \hat{\pi})^{T-x} \hat{\pi}^x.$$

The likelihood-ratio statistic is

$$LR_{uc} = -2 \ln \left(\frac{L_0}{L_1} \right),$$

which is asymptotically χ_1^2 under the null.

5.9 Christoffersen independence and conditional coverage

Let n_{ij} count transitions from state i at time $t-1$ to state j at time t , where $i, j \in \{0, 1\}$. Under independence, the one-step transition probability into breach is the same regardless of the previous state. Under the alternative, use separate probabilities

$$\pi_0 = \frac{n_{01}}{n_{00} + n_{01}}, \quad \pi_1 = \frac{n_{11}}{n_{10} + n_{11}}.$$

Comparing the null and alternative Markov likelihoods produces the independence statistic LR_{ind} . Adding Kupiec's unconditional-coverage statistic gives

$$LR_{cc} = LR_{uc} + LR_{ind},$$

which is asymptotically χ_2^2 . That is why a model can pass LR_{uc} but fail LR_{cc} : the count may be right while the timing pattern is wrong.

5.10 Why stress numbers exceed ordinary daily VaR

A full-sample VaR compresses many regimes into one average tail description. A crisis-window stress test isolates exactly the regimes where correlation rises, liquidity worsens, and volatility explodes. That is why the COVID stress-window 99% VaR in the notebook is almost three times the baseline full-sample 99% VaR. The two numbers answer different questions, and both are needed in practice.